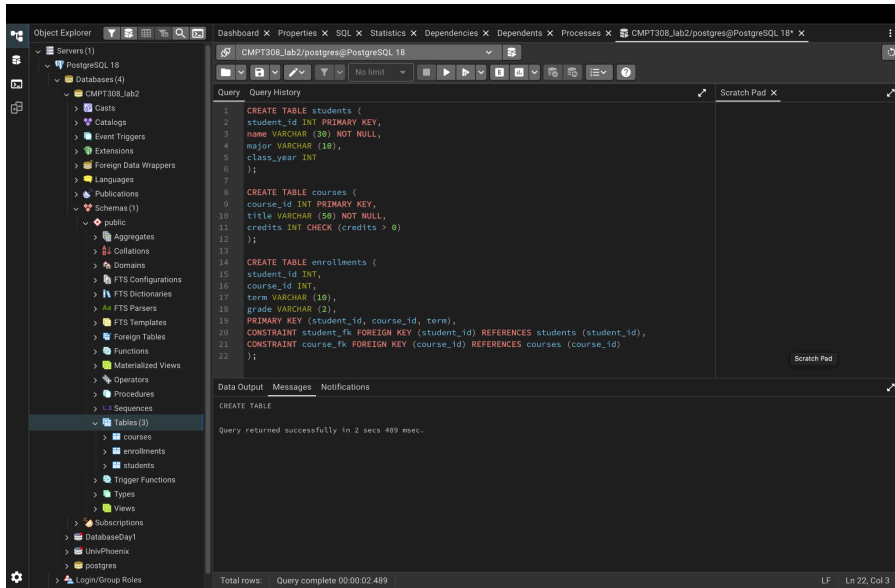


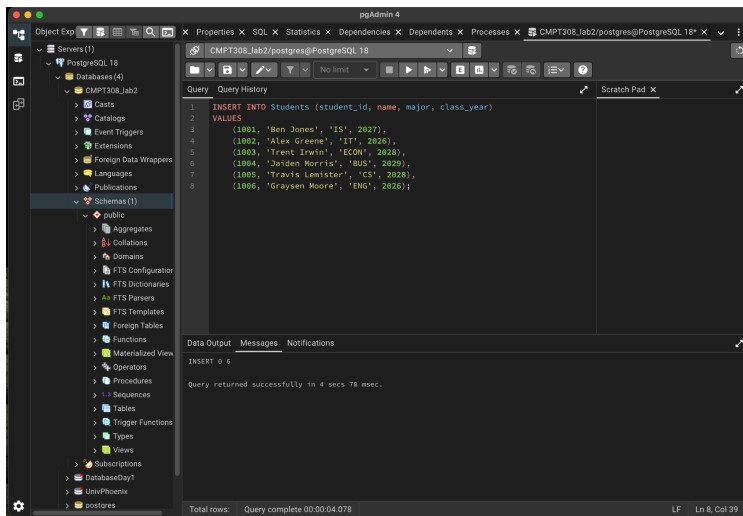
Part C:



The screenshot shows the pgAdmin 4 interface with the 'Tables (3)' folder selected in the left pane. The main query editor displays the following SQL script:

```
1 CREATE TABLE students (  
2   student_id INT PRIMARY KEY,  
3   name VARCHAR (30) NOT NULL,  
4   major VARCHAR (10),  
5   class_year INT  
6 );  
7  
8 CREATE TABLE courses (  
9   course_id INT PRIMARY KEY,  
10  title VARCHAR (50) NOT NULL,  
11  credits INT CHECK (credits > 0)  
12 );  
13  
14 CREATE TABLE enrollments (  
15   student_id INT,  
16   course_id INT,  
17   term VARCHAR (10),  
18   grade VARCHAR (2),  
19   PRIMARY KEY (student_id, course_id, term),  
20   CONSTRAINT student_fk FOREIGN KEY (student_id) REFERENCES students (student_id),  
21   CONSTRAINT course_fk FOREIGN KEY (course_id) REFERENCES courses (course_id)  
22 );
```

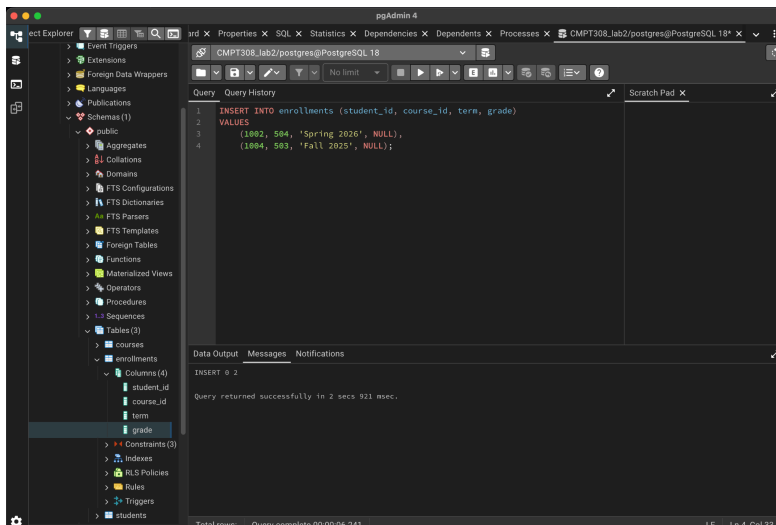
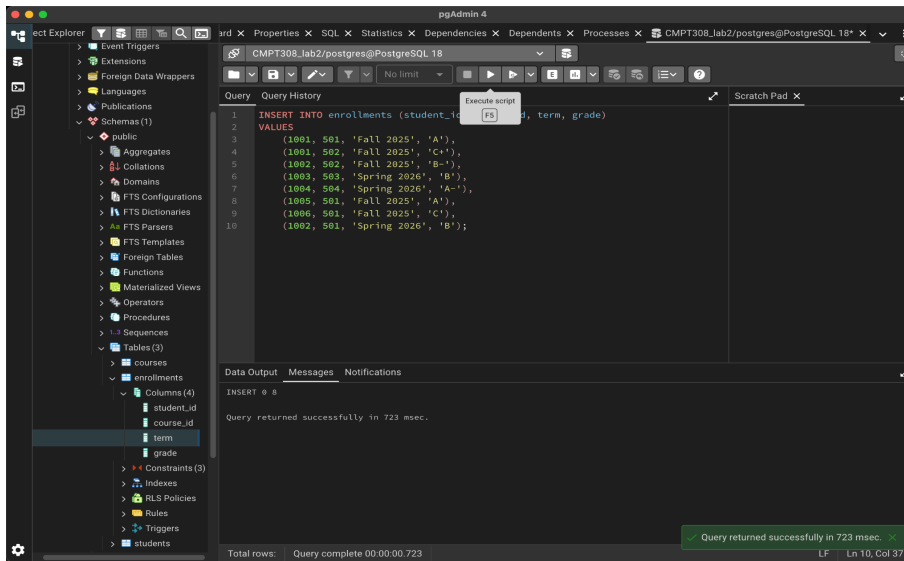
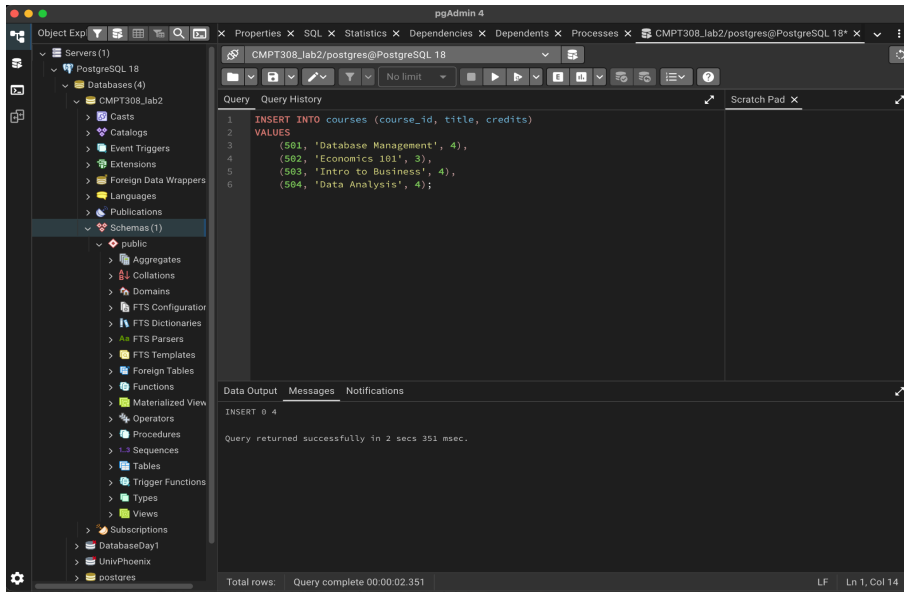
The 'Data Output' pane shows the message: 'Query returned successfully in 2 secs 489 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:02.489'.



The screenshot shows the pgAdmin 4 interface with the 'Tables (3)' folder selected in the left pane. The main query editor displays the following SQL script:

```
1 INSERT INTO students (student_id, name, major, class_year)  
2 VALUES  
3   (1001, 'Ben Jones', 'IS', 2027),  
4   (1002, 'Ales Greene', 'IT', 2026),  
5   (1003, 'Trent Irwin', 'ECON', 2026),  
6   (1004, 'Jaiden Morris', 'BUS', 2025),  
7   (1005, 'Travis Lemister', 'CS', 2026),  
8   (1006, 'Grayson Moore', 'ENG', 2026);
```

The 'Data Output' pane shows the message: 'Query returned successfully in 4 secs 78 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:04.078'.



pgAdmin 4

CMPT308_lab2/postgres@PostgreSQL 18*

Query: `SELECT * FROM courses`

Data Output: Messages Notifications

course_id [PK] integer	title character varying (50)	credits integer
1	501 Database Management	4
2	502 Economics 101	3
3	503 Intro to Business	4
4	504 Data Analysis	4

Total rows: 4 Query complete 00:00:13.238 LF Ln 1, Col 22

pgAdmin 4

CMPT308_lab2/postgres@PostgreSQL 18*

Query: `SELECT * FROM students`

Data Output: Messages Notifications

student_id [PK] integer	name character varying (30)	major character varying (10)	class_year integer
1	1001 Ben Jones	IS	2027
2	1002 Alex Greene	IT	2026
3	1003 Trent Irwin	ECON	2028
4	1004 Jaden Morris	BUS	2029
5	1005 Travis Lemister	CS	2028
6	1006 Grayson Moore	ENG	2026

Total rows: 6 Query complete 00:00:03.314 LF Ln 1, Col 23

pgAdmin 4

CMPT308_lab2/postgres@PostgreSQL 18*

Query: `SELECT * FROM enrollments`

Data Output: Messages Notifications

student_id [PK] integer	course_id [PK] integer	term [PK] character varying (20)	grade character varying (2)
1	1001	501 Fall 2025	A
2	1001	502 Fall 2025	C+
3	1002	502 Fall 2025	B-
4	1003	503 Spring 2026	B
5	1004	504 Spring 2026	A-
6	1005	501 Fall 2025	A
7	1006	501 Fall 2025	C
8	1002	501 Spring 2026	B
9	1002	504 Spring 2026	[null]
10	1004	503 Fall 2025	[null]

Total rows: 10 Query complete 00:00:05.866 LF Ln 1, Col 26

pgAdmin 4

Object Explorer: PostgreSQL 18, CMPT308_lab2, public schema.

Query: `SELECT * FROM students WHERE major = 'ECON';`

Data Output:

student_id [PK] integer	name character varying (30)	major character varying (10)	class_year integer	
1	1003	Trent Irwin	ECON	2028

Query: `SELECT student_id, name FROM students`

Data Output:

student_id [PK] integer	name character varying (30)	
1	1001	Ben Jones
2	1002	Alex Greene
3	1003	Trent Irwin
4	1004	Jaiden Morris
5	1005	Travis Lemister
6	1006	Graysen Moore

pgAdmin 4

Object Explorer: PostgreSQL 18, CMPT308_lab2, public schema.

Query: `SELECT * FROM students WHERE name LIKE '%T%';`

Data Output:

student_id [PK] integer	name character varying (30)	major character varying (10)	class_year integer	
1	1003	Trent Irwin	ECON	2028
2	1005	Travis Lemister	CS	2028

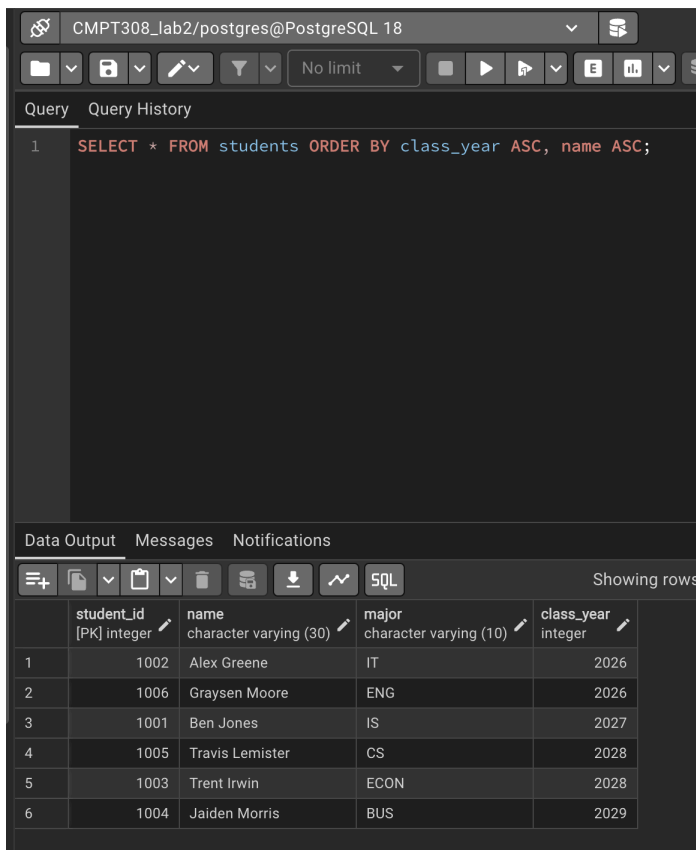
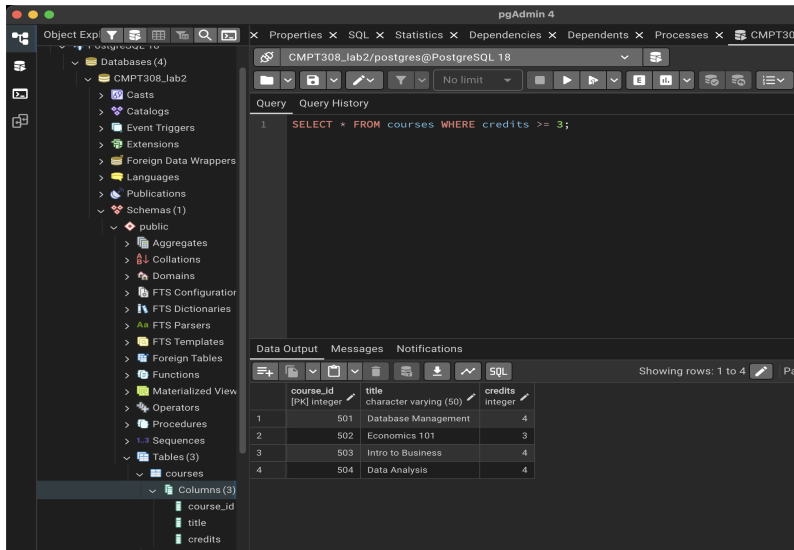
pgAdmin 4

Object Explorer: PostgreSQL 18, CMPT308_lab2, public schema.

Query: `SELECT * FROM enrollments WHERE grade IS NULL;`

Data Output:

student_id [PK] integer	course_id [PK] integer	term [PK] character varying (20)	grade character varying (2)	
1	1002	504	Spring 2026	[null]
2	1004	503	Fall 2025	[null]



Part D:

Select students whose major is a value you used.

$\sigma_{\text{major} = \text{'Econ'}}(\text{Students})$

List only student-id and name from (Students)

$\pi_{\text{student-id}, \text{name}}(\text{Students})$

https://github.com/TravisLemister/Database_Management_labs/blob/main/CMPT308_lab2